EE2004: Digital Signal Processing

Tutorial 2 Answer Key

1. (a) $\sin(0.4\pi n - \theta \pm \pi)$, (b) $\cos(0.3\pi n + \theta)$ (c) $\cos(0.9549\pi n - \theta)$, (d) $\sin(0.3\pi n - \theta \pm \pi)$
2. (a) $\omega_1 = 0$, (b) $\omega_1 = \pi$, (c) $\omega_1 = \pi$, (d) $\omega_1 = 0.8\pi$ (phase changes sign), (e) $\omega_1 = 0.9549\pi$ (phase changes sign), (f) $\omega_1 = 0.3634\pi$ (phase changes sign)
3. $\theta = \omega_0 N \pm 2\pi k$
4. 0
5. - $\omega \in [-\pi, \pi]$
   (a) $\omega = 0.2\pi$, θ, (b) $\omega = 0$, 0, (c) $\omega = -1.95$, 0, (d) $\omega = -0.7\pi$, 0.
   - $\omega \in [0, 2\pi)$
   (a) and (b) are same as before, (c) $\omega = 2\pi - 1.95$, 0, (d) $\omega = 1.3\pi$, 0.
6. (a) 1, (b) 20, (c) 10, (d) 5, (e) 20, (f) 2, (g) 20, (h) 5, (i) not periodic.
7. 0.16π, 0.24π.
8. (a) $N = 11$ for $k = 1$ to 5, (b) the (k, N) pairs for $k = 1$ to 6 are $(1, 12)$, $(2, 6)$, $(3, 4)$, $(4, 3)$, $(5, 12)$, $(6, 2)$. If $k = 0$, the signal is DC whose period is 1.
9. (i) non-memoryless, time-variant, linear, non-causal, stable; (ii) non-memoryless, time-variant, linear, non-causal, stable; (v) non-memoryless, time-variant, linear, non-causal, stable.
10. (a) non-linear, time-invariant, (b) linear, time-variant, (c) incrementally linear time-variant.
11. (a) linear, causal, time-invariant (b) non-linear, time-variant, causal, (c) linear, time-variant, causal, (d) non-linear, time-invariant, causal, (e) incrementally linear, time-variant, causal, (f) non-linear, time-invariant, causal, (g) non-linear, time-invariant, causal, (h) non-linear, time-invariant, non-causal, (i) non-linear, time-invariant, causal.
12. $\dfrac{1}{a^{-1} - b}\, a^{-n-1}\, u[n] + \dfrac{1}{a^{-1} - b}\, b^{n+1}\, u[-n-1]$
13. Proof.
14. (a) $(n+1)\, u[n]$, (b) does not exist.
15. (a) $\phi_{xw}[k] = x[k] * w[-k]$, (b) $\phi_{xw}[k] = \phi_{wx}[-k])$, $\phi_{xx}[k] = \phi_{xx}[-k]$, (c) shifted triangle, triangle centred at $k = 0$.
16. (a) $y[n] = 0.8^n\, u[n]$, (b) When $x[n] = 0.5\, u[n]$, output is same as $0.5\, u[n] * 0.8^n\, u[n]$.

17.

$$\mathbf{X} = \begin{bmatrix} x[0] & x[-1] & x[-2] & ... & x[-N+2] & x[-N+1] \\ x[1] & x[0] & x[-1] & ... & x[-N+3] & x[-N+2] \\ x[2] & x[1] & x[0] & ... & x[-N+4] & x[-N+3] \\ . & . & . & ... & . & . \\ . & . & . & ... & . & . \\ x[N-1] & x[N-2] & x[N-3] & ... & x[1] & x[0] \end{bmatrix}$$

$$y = \begin{bmatrix} y[0] & y[1] & y[2] & ... & y[N-1] \end{bmatrix}^T$$

$$h = \begin{bmatrix} h[0] & h[1] & h[2] & ... & h[N-1] \end{bmatrix}^T$$

18. $h = \{\overset{\downarrow}{1}, 1, 2\}$

19. $\{\overset{\downarrow}{1}, 3, 1\} * \{\overset{\downarrow}{1}, 2, 2\} = \{\overset{\downarrow}{1}, 5, 9, 8, 2\}$. $131 \times 122 = 15982$.

20.

$$y[n] = \begin{cases} 0 & \text{if } n < 0, \\ 2 - 2^{-n} & \text{if } 0 \leq n \leq 9, \\ 2^{-n}(2^{10} - 1) & \text{if } n \geq 10 \end{cases}$$